Employee Attrition Analysis Project

Data Cleaning Report

1. Dataset Overview

The dataset (WA_Fn-UseC_-HR-Employee-Attrition.csv) contains 1,470 employee records across 35 attributes, combining demographic, compensation, satisfaction, and tenure information along with the target variable, Attrition.

Metric	Value
Total records	1,470
Total columns (raw)	35
Missing values	0 (across all columns)
Duplicate rows	0
Duplicate EmployeeNumber (ID) values	0 — confirmed unique primary key
Target variable	Attrition (Yes / No)

2. Data Dictionary

Below is the full data dictionary for the raw dataset, documenting each field, its type, and its meaning.

Column	Type	Description
Age	Numeric	Employee age in years
Attrition	Categorical (Yes/No)	Target variable — whether the employee left the company
BusinessTravel	Categorical	Frequency of business travel (Non-Travel / Travel_Rarely / Travel_Frequently)
DailyRate	Numeric	Daily pay rate
Department	Categorical	Department the employee belongs to
DistanceFromHome	Numeric	Distance from home to office (km)
Education	Ordinal (1-5)	1=Below College, 2=College, 3=Bachelor, 4=Master, 5=Doctor
EducationField	Categorical	Field of study
EmployeeNumber	Numeric (ID)	Unique employee identifier
EnvironmentSatisfaction	Ordinal (1-4)	1=Low, 2=Medium, 3=High, 4=Very High
Gender	Categorical	Male / Female
HourlyRate	Numeric	Hourly pay rate
JobInvolvement	Ordinal (1-4)	1=Low, 2=Medium, 3=High, 4=Very High
JobLevel	Ordinal (1-5)	Seniority level within the company
JobRole	Categorical	Specific job title
JobSatisfaction	Ordinal (1-4)	1=Low, 2=Medium, 3=High, 4=Very High
MaritalStatus	Categorical	Single / Married / Divorced
MonthlyIncome	Numeric	Monthly salary (currency units)
MonthlyRate	Numeric	Monthly pay rate (contractual)
NumCompaniesWorked	Numeric	Number of companies worked at previously
OverTime	Categorical (Yes/No)	Whether the employee regularly works overtime
PercentSalaryHike	Numeric	Percentage salary increase in the last review
PerformanceRating	Ordinal (1-4)	3=Excellent, 4=Outstanding (only these two appear in the data)
RelationshipSatisfaction	Ordinal (1-4)	1=Low, 2=Medium, 3=High, 4=Very High
StockOptionLevel	Ordinal (0-3)	Level of stock options granted
TotalWorkingYears	Numeric	Total years of professional experience
TrainingTimesLastYear	Numeric	Number of training sessions attended last year
WorkLifeBalance	Ordinal (1-4)	1=Low, 2=Good/Medium, 3=High, 4=Very High
YearsAtCompany	Numeric	Tenure at the current company
YearsInCurrentRole	Numeric	Years spent in the current role
YearsSinceLastPromotion	Numeric	Years since the last promotion
YearsWithCurrManager	Numeric	Years working with the current manager

3. Data Quality Checks Performed

- Checked for missing/null values across all 35 columns — none found.
- Checked for fully duplicated rows — none found.
- Verified EmployeeNumber is unique for every record (valid primary key).
- Verified categorical fields (Attrition, Gender, OverTime, Department, etc.) contain only expected, consistently-spelled category values — no typos or case-mismatches found.
- Checked data types — all numeric fields load correctly as integers; no columns needed type coercion.
- Scanned for constant (zero-variance) columns that carry no analytical signal.
- Ran an IQR-based outlier scan on 14 key numeric fields to distinguish genuine data errors from legitimate skew.

4. Columns Removed

Three columns were constant across every one of the 1,470 records and were removed as they add no analytical or predictive value:

Column	Observation	Action
EmployeeCount	Constant value (1) for every record	No analytical value; removed
Over18	Constant value ("Y") for every record	No analytical value; removed
StandardHours	Constant value (80) for every record	No analytical value; removed

Resulting cleaned dataset: 1,470 rows × 32 core columns (plus 6 derived label columns described in Section 6).

5. Outlier Assessment

An IQR (1.5× interquartile range) scan was run on the key continuous numeric fields. Flagged values were reviewed individually rather than removed automatically, since attrition analysis specifically depends on understanding extreme/skewed behavior (e.g. employees not promoted in years, or very new hires).

Field	Flagged as outlier (IQR)	Cleaning decision
MonthlyIncome	114 (7.8%)	Right-skewed pay structure across job levels — expected, not an error
NumCompaniesWorked	52 (3.5%)	A few employees with many prior employers — retained (valid)
TotalWorkingYears	63 (4.3%)	Senior/long-tenured staff — retained (valid)
TrainingTimesLastYear	238 (16.2%)	Discrete count variable (0-6); "outliers" are just less common values — retained
YearsAtCompany	104 (7.1%)	Long-tenured employees — retained (valid)
YearsInCurrentRole	21 (1.4%)	Retained (valid)
YearsSinceLastPromotion	107 (7.3%)	Employees not promoted in a long time — retained (this is analytically important, not noise)
YearsWithCurrManager	14 (1.0%)	Retained (valid)

Decision: No rows were dropped for outliers. All flagged values are legitimate real-world data points (e.g., senior employees, long-since-promoted staff) that are directly relevant to attrition analysis. Removing them would bias the study toward "typical" employees and hide exactly the patterns the project needs to find.

6. Feature Engineering for Readability

The dataset encodes several satisfaction/rating fields as integers (1-4). To make charts and dashboards interpretable for a non-technical HR audience, corresponding text-label columns were added (original numeric columns were retained for statistical calculations):

- EnvironmentSatisfaction_Label, JobSatisfaction_Label, RelationshipSatisfaction_Label, WorkLifeBalance_Label, JobInvolvement_Label → Low / Medium / High / Very High
- Education_Label → Below College / College / Bachelor / Master / Doctor
- PerformanceRating_Label → Low / Good / Excellent / Outstanding

- AttritionFlag (0/1) was added as a numeric encoding of Attrition, used for correlation and rate calculations.

7. Final Cleaned Dataset

Metric	Value
Final rows	1,470
Final columns	39 (32 core + 6 readability labels + 1 attrition flag)
Missing values remaining	0
Ready for analysis / dashboarding	Yes

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